

Root Finding: Summary

Bracketing, Newton, and how to combine them

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Bracketing methods: bisection and false position

Start from a **bracket** $[a, b]$: f continuous, $f(a)$ and $f(b)$ of opposite sign. Pick a point c inside, keep the half with the sign change.

Bisection: midpoint

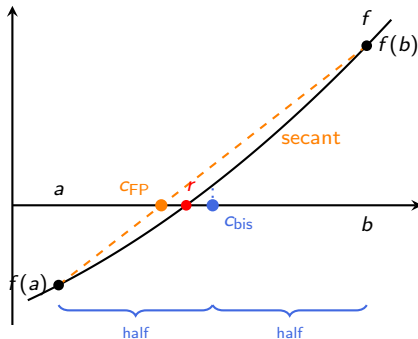
$$c = a + \frac{1}{2}(b - a)$$

Uses only the **sign** of f .

False position: zero of the secant

$$c = a + \frac{-f(a)}{f(b) - f(a)}(b - a)$$

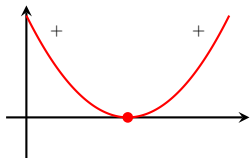
The weight $\frac{1}{2}$ is replaced by a fraction in $(0, 1)$ set by the **values** of f : often faster, but one endpoint can get stuck.



Same bracket, two choices of c : **midpoint** or **secant zero**.

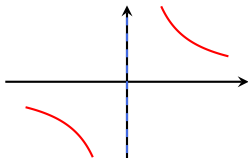
When can bisection go wrong?

Both conditions of the IVT matter (continuity and a sign change). Always **plot f first**.



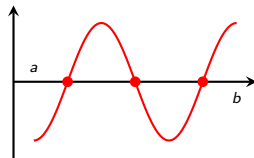
Root, no sign change

$(x - 1)^2$ touches the axis but never changes sign: no bracket exists, so the root cannot be found.



Sign change, no root

$1/x$ on $[-1, 1]$ is not continuous. Bisection “converges” to the jump at $x = 0$.



Several roots

Still a sign change, but you get *one* root, and which one depends on the bracket.

Also watch for

Tolerance below machine precision ($\text{tol} < 10^{-16}|r|$): the midpoint equals a or b and the loop never ends, so always cap with `maxit`. **False position:** a stuck endpoint can make it much *slower* than bisection.

Bisection: guaranteed, linear, and predictable

- ▶ **Guaranteed:** the bracket always contains a root.
- ▶ **Linear,** $\rho = \frac{1}{2}$: the bound halves every step.

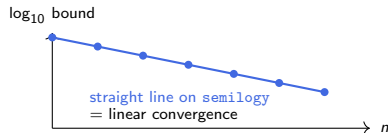
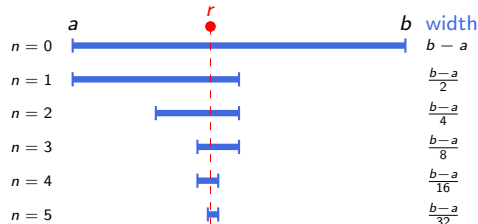
Steps from the tolerance

$$|c_n - r| \leq \frac{b-a}{2^n} \leq \text{tol} \iff n \geq \log_2 \frac{b-a}{\text{tol}}$$

r is in the bracket and c_n is its midpoint:
 $|c_n - r| \leq$ half its width. Known **before** we start.

$[a, b] = [1, 2], \text{ tol}$	10^{-3}	10^{-6}	10^{-12}
steps n	10	20	40

Cost = $O(\log(1/\text{tol}))$: $10\times$ accuracy ≈ 3.3 more steps.



Newton's method: fast, but only local

Follow the **tangent line** at x_n to the axis:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Quadratic convergence

$e_{n+1} \approx \frac{f''(r)}{2f'(r)} e_n^2$: correct digits **double** each step.

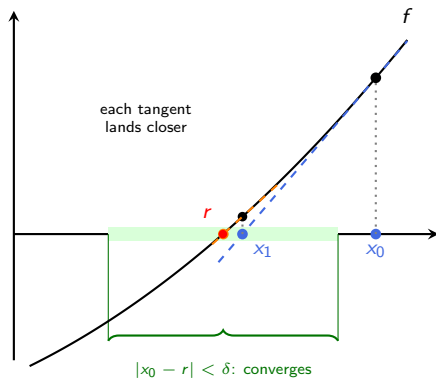
If $|e_n| \approx 10^{-k}$, then $|e_{n+1}| \approx C \cdot 10^{-2k}$.

Local: no guarantee

Converges only if x_0 is **close enough**

($|x_0 - r| < \delta$, with δ unknown) and $f'(r) \neq 0$.

No bracket, no error bound in advance.



Newton without f' : finite differences and secant

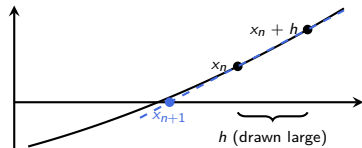
Newton only needs **a slope** at x_n . Choose the variation by what you have:

f is a function, but f' is not given

Finite-difference Newton: evaluate f at a nearby point

$$f'(x_n) \approx \frac{f(x_n + h) - f(x_n)}{h}, \quad h \approx 10^{-8} |x_n|$$

Each step costs **2 evaluations of f** : $f(x_n)$ and $f(x_n + h)$. Nearly quadratic. h too big: slope error; too small: rounding.

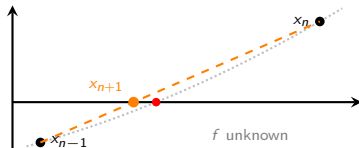


Only a handful of points (x_k, f_k)

Secant method: slope through the **two newest** points

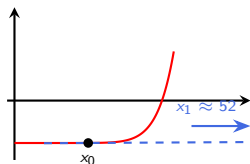
$$x_{n+1} = x_n - f_n \frac{x_n - x_{n-1}}{f_n - f_{n-1}}$$

No f' , no bracket. Each step costs **1 evaluation of f** , at the new point x_{n+1} ; f_{n-1} is reused. Order ≈ 1.618 .



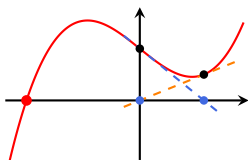
When can Newton (or secant) go wrong?

Only *local* information: far from the root, the tangent (or secant) can point anywhere.



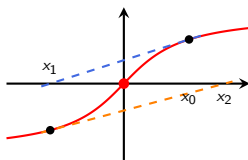
Flat tangent

$f'(x_n) \approx 0$: huge overshoot.
 $f'(x_n) = 0$: step undefined.



Cycling

$x^3 - 2x + 2$ from $x_0 = 0$:
0, 1, 0, 1, ... forever.



Divergence

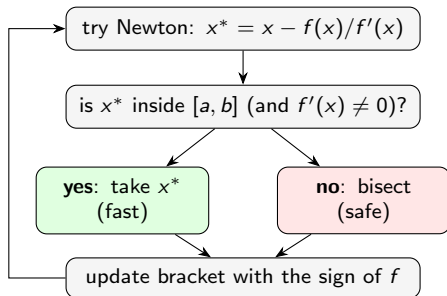
$\arctan x$ from $x_0 = 1.5$:
 $-1.69, 2.32, -5.11, \dots$

Defences

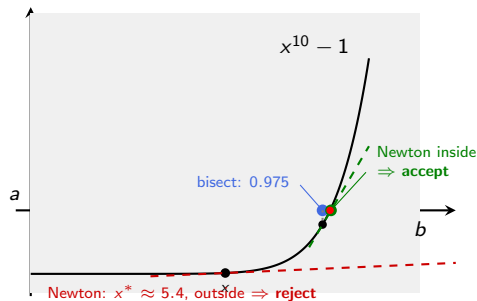
Start close (plot f first) · cap with `maxit` and **raise** if reached · a small step is not a root: also check $|f(x)|$ · secant: $f_n = f_{n-1}$ gives division by zero.

Best of both: bisection + Newton (safeguarded Newton)

Keep a bracket $[a, b]$ at all times. Each step:



Guaranteed like bisection; **quadratic** near the root like Newton.



$x^{10} - 1, [0, 1.3]$	bisect.	false pos.	Illinois	Newton	safeguarded
steps to 10^{-6}	20	61	13	21	4